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Problem 1.3.A.9:

$\tilde{r}_{nc}(f) = N_0/2\,(\mathrm{rect}_{b_1}(f-f_c) + \mathrm{rect}_{b_1}(-f-f_c))$

$b_1 = 7\,MHz$

$\tilde{h}_L = \mathrm{rect}_b(f),\ b = 5MHz$

1) $\tilde{r}_n(f) = 2\tilde{r}_{nc}(f+f_c)|\tilde{h}_L(f)|^2$

$= N_0(\mathrm{rect}_{b_1}(f) + \mathrm{rect}_{b_1}(-f))\,|\mathrm{rect}_b(f)|^2$

The $\mathrm{rect}_b(f)$ term is a smaller window since $b < b_1$ so:

$\tilde{r}_n(f) = N_0\,\mathrm{rect}_b(f)$

$Var[n(t)] = \int_{-\infty}^{\infty} \tilde{r}_n(f)\,df$

$= \int_{-\infty}^{\infty} N_0\,\mathrm{rect}_b(f)\,df$

Area of $\mathrm{rect}_b = b$

$$\boxed{Var[n(t)] = N_0 b = N_0\ 7\times10^7}$$

2) $f_n(n) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-|n-\mu|^2}{2\sigma^2}}$, for noise $\mu = \emptyset$ and from pt 1, $\sigma^2 = N_0 b$

$$\boxed{f_n(n) = \frac{1}{\sqrt{2N_0 b\pi}} e^{\frac{-n^2}{2N_0 b}}}$$

3) $f_{y(t_0)}(y) = \frac{1}{\sqrt{2\pi\sigma^2_{y(t_0)}}} e^{\frac{-|y-\mu|^2}{2\sigma^2_{(y(t_0))}}}$

$y(t_0) = n(t_0) - 1.5$

$\mu_{y(t_0)} = \mu_{n(t_0)} + \mu_{x(t_0)}$

$= \emptyset - 1.5$

$= -1.5$

$Var[y(t_0)] = Var[x(t_0) + n(t_0)]$

$= Var[x(t_0)] + Var[n(t_0)]$

$= \emptyset + N_0 b$

$= N_0 b$

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Problem 1.3.A.9 (cont.):

3) $f_{y(t_0)}(y) = \frac{1}{\sqrt{2\pi\sigma^2_{y(t_0)}}}\left(e^{\frac{-|y-N_{y(t_0)}|^2}{2N_{y(t_0)}}}\right)$

$$\boxed{f_{y(t_0)}(y) = \frac{1}{\sqrt{2\pi N_0 b}}\left(e^{\frac{-|y+1.5|^2}{2N_0 b}}\right)}$$

4) $r_n(\tau) = \mathcal{F}^{-1}\{\tilde{r}_n(f)\}(\tau)$

$\tilde{r}_n(f) = N_0\, rect_b(f)$ From part 1

$\mathcal{F}^{-1}\{N_0\, rect_b(f)\} = N_0 b\, sinc(tb)$

$r_n(\tau) = N_0 b\, sinc(\tau b)$

$E[(n(t_1)-E[n(t_1)])(n(t_2)-E[n(t_2)])^*] = r_n(t_1-t_2)$

$= r_n(\tau)$

$Var[n(t)] = N_0 b$

$Var[n(t_1)] = Var[n(t_2)] = N_0 b$

$\rho = \frac{r_n(\tau)}{Var[n(t)]} = \frac{N_0 b\, sinc(\tau b)}{N_0 b}$

$\boxed{\rho = sinc(\tau b)} = sinc((-3\times10^{-8})(5\times10^{6})) = \boxed{0.963}$

5) For independence, $r_n(\tau)$ must be $\emptyset$

$t_3 = 5\times10^{-7}$, $t_4 < t_5$, $r_n(t_4-t_3) = \emptyset$, $r_n(t_5-t_3) = \emptyset$

$r_n(\tau) = N_0 b\, sinc(\tau b)$

$= \frac{N_0 \sin(\pi b \tau)}{\tau}$

$\sin(\pi \tau b) = \emptyset \Rightarrow r_n(\tau) = \emptyset$ if $\tau \neq \emptyset$

Analyzing the zero crossings of $\sin(\pi(t-5\times10^{-7})(7\times10^{6}))$:

$\cancel{t = \frac{1}{7\times10^6} + 5\times10^{-7}}$ $\cancel{t = \frac{1}{7\times10^6} + 5\times10^{-7}}$

$\cancel{= 7.5\times10^{-7}}$ $\cancel{= 1.42\times10^{-7} + 5\times10^{-7} = 6.42\times10^{-7}}$

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Problem 1.3.A.9 (cont):

$\sin(\pi(t-5\times10^{-7})(7\times10^{6}))$ is $\emptyset$ when $(t-5\times10^{-7})(7\times10^{6}) = \emptyset$

$$\boxed{\begin{aligned} t_4 &= 3.57\times10^{-7} \\ t_5 &= 6.43\times10^{-7} \end{aligned}}$$

6) $r_{nI}(\tau) = \frac{1}{2}\,\mathrm{Re}[r_n(\tau)]$

$= \frac{1}{2}\,\mathrm{Re}[N_0 b\,\mathrm{sinc}(b\tau)]$

$= \frac{1}{2} N_0 b\,\mathrm{sinc}(b\tau)$ since sinc is real-valued

$E[(n_I(t_1) - E[n_I(t_1)])(n_I(t_2) - E[n_I(t_2)])]$ is what we want to find first

$E[n_I(t_1)] = \emptyset$ for all t

$E[(n_I(t_1))(n_I(t_2))] = r_{nI}(\tau) = \frac{1}{2} N_0 b\,\mathrm{sinc}(b\tau)$

$\mathrm{Var}[n_I(t_1)] = \int_{-\infty}^{\infty} F\{n_I(t)\}\,df$

$= \int_{-\infty}^{\infty} \frac{1}{2} N_0\,\mathrm{rect}_b\,df$

$= \frac{1}{2} N_0 b$ for any value of t

$\mathrm{Var}[n_I(t_1)] = \mathrm{Var}[n_I(t_2)]$

$$\rho = \frac{\frac{1}{2} N_0 b\,\mathrm{sinc}(b\tau)}{\frac{1}{2} N_0 b}$$

$$\boxed{\rho = \mathrm{sinc}(b\tau) = 0.93}$$

Problem 1.4.A.1:

$|a| = \frac{1}{\sqrt{4\pi r^2}}$, $r = 100\,m$

1) $|a| = \frac{1}{\sqrt{4\pi(100)^2}} = \boxed{\frac{1}{\sqrt{40000\pi}}}$

2) $\tau = r/c$

$= 100/c \approx \boxed{334\,ns}$

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Problem 1.A.4.2:

$a = \frac{1}{\sqrt{4\pi r^2}} e^{-j\pi/5}$, $x(t) = e^{j(2\pi f_0 t + \theta_0)}$, $r = 80m$, $f_0 = 1kHz$, $\theta_0 = \pi/3$, $f_c = 2GHz$

1) $a = \frac{1}{\sqrt{25600\pi}} e^{-j\pi/5}$

$|a| = \frac{1}{\sqrt{26600\pi}}$

$20\log_{10}|a| = \boxed{-49.05\,dB}$

2) $\tau = r/c = \boxed{266.9ns}$

3) $y_c(t) = Re[a e^{-j2\pi f_c \tau} e^{j2\pi f_1 (t-\tau)} \sqrt{2} e^{j2\pi f_c t}]$

$= |a| \cos(-2\pi f_c \tau) \cos(2\pi f_1 (t-\tau)) \sqrt{2} \cos(2\pi f_c t)$

$\boxed{y_c(t) = 4.99\times10^{-3} \cos(2000\pi(t - 266.9\times10^{-9})) \cos(2\times10^9\, 2\pi f_c)}$

Problem 1.A.4.3:

$x(t) = e^{j2\pi f_1 t}$, $f_c = 2GHz$, $t_b = \frac{1}{5\times10^6}$ s/symbol

1) $f_d = \frac{-r'(t_0)}{\lambda_{f_c}}$

$\lambda_{f_c} = c/f_c = c/2GHz \approx 0.150m$

65mph ≈ 29.06 m/s $= r'(t)$

$\boxed{f_d = -193.73\ Hz}$

2) $f_d t_b = \frac{-r'(t_0)}{\lambda_f \frac{1}{t_b}}$ $t_b = 1/5\times10^6$

$\boxed{f_d t_b = 38.75\times10^{-6}}$

3) $f_{de} = (1 + f_1/f_c) f_d = (1 + 2.5\times10^6/2\times10^9)(-193.73) = \boxed{-193.97\ Hz}$

4) $f_{de} = (1 + f_1/f_c) f_d = (1 - 2.5\times10^6/2\times10^9)(-193.73) = \boxed{-193.49\ Hz}$